

CSIT213 Java Programming
Assignment 1 (15%)
Due: refer to Moodle

Objectives

This assignment is designed to help students strengthen their Java foundational through a wide range of programming problems. By the end of this assignment, students should be able to:

- Strengthen Core Java Programming Skills
- Apply Algorithmic Thinking
- Practice Fundamental Algorithms
- Work With Data Structures
- Develop Problem-Solving and Debugging Skills

Tasks

Refer to A1.java. Each task is an individual programming problem students must solve by writing a function.

- Q1 - Write a function to return the sum of all odd numbers in the int array
- Q2 - Write a function to return the largest value in the int array
- Q3 - Write a function to return a new int array that is the reverse of the input array
- Q4 - Write a function to return the index of target in the array, or -1 if it is not found (linear search)
- Q5 - Write a function to return the product of all elements in the int array
- Q6 - Write a function to return how many times target appears in the array
- Q7 - Write a function to return a new int array where every element is doubled
- Q8 - Write a function to return the second largest distinct value in the array
- Q9 - Write a function to return the range of the array (largest value minus smallest value)
- Q10 - Write a function to return a new int array sorted in ascending order
- Q11 - Write a function to return the number of distinct (unique) values in the array
- Q12 - Write a function to return a new int array rotated one place to the right
- Q13 - Write a function to return the number of vowels (a, e, i, o, u) in the string
- Q14 - Write a function to return the reverse of the string
- Q15 - Write a function to return true if the string is a palindrome, otherwise false
- Q16 - Write a function to return the string with every space replaced by an underscore '_'
- Q17 - Write a function to return the number of uppercase letters in the string
- Q18 - Write a function to return the string with all vowels removed
- Q19 - Write a function to return true if the two strings are anagrams of each other, otherwise false
- Q20 - Write a function to return the first character that does not repeat in the string
- Q21 - Write a function to return the string in title case
- Q22 - Write a function to return the character that appears most often in the string
- Q23 - Write a function to return how many times the substring sub appears in s (non-overlapping)
- Q24 - Write a function to return true if the string contains only digit characters (0-9), otherwise false
- Q25 - Write a function to return the string with duplicate characters removed, keeping the first occurrence

- Q26 - Write a function to return true if n is a prime number, otherwise false
- Q27 - Write a function to return the factorial of n ($n! = 1 * 2 * 3 * \dots * n$). Assume $n \geq 0$.
- Q28 - Write a function to return the nth Fibonacci number where $\text{fib}(0)=0$, $\text{fib}(1)=1$
- Q29 - Write a function to return the sum of the digits of a positive integer n
- Q30 - Write a function to return how many digits are in a positive integer n
- Q31 - Write a function to return the integer formed by reversing the digits of a positive integer n
- Q32 - Write a function to return true if year is a leap year, otherwise false
- Q33 - Write a function to return the greatest common divisor (GCD) of two positive integers a and b
- Q34 - Write a function to return how many multiples of 3 there are from 1 to n (inclusive)
- Q35 - Write a function that takes an int array and returns an `ArrayList<Integer>` containing only the even numbers
- Q36 - Write a function to return the sum of all numbers in an `ArrayList<Integer>`
- Q37 - Write a function to return a new `ArrayList<Integer>` with duplicates removed, keeping the original order
- Q38 - Write a function to return a new `ArrayList<Integer>` containing only the elements greater than threshold
- Q39 - Write a function to return the largest number in an `ArrayList<Integer>`
- Q40 - Write a function to return the sum of all elements in a 2D int array (a grid)
- Q41 - Write a function to return the sum of all elements in a given row of the grid
- Q42 - Write a function to return the sum of all elements in a given column of the grid
- Q43 - Write a function to return the largest value in the whole 2D array
- Q44 - Write a function to return the sum of the main diagonal of a square grid
- Q45 - Write a function to return the sum of the anti-diagonal of a square grid
- Q46 - Write a function to return how many times target appears in the 2D array
- Q47 - Write a function to return an int array where each element is the sum of that row in the grid
- Q48 - Write a function to return an int array where each element is the largest value in that row
- Q49 - Write a function to return the transpose of a rectangular grid
- Q50 - Write a function to return true if the grid is rectangular (every row has the same length)

Test Code

- The testing code **A1_Assertion.java** is provided to assist you to test your code. You need to enable assertion in Java to enable the testing.
- The first 10 test cases are given for your reference. You need to work on the rest to test your codes.
- Do not import extra packages (except for ArrayList) for the Java code

Submission:

- **Do not use any packages for the Java code.**
- Failure to compile and execute the code will result in **zero marks awarded**.
- **Copy your Java file to a folder and name it using your tutorial group, name and UOW id.**
- For full time student, the zip file name must be in the form of: TXXF_NAME_UOWID.zip where XX is your tutorial group, NAME is your full name (without space and underscore) and UOW is your 7-digit UOW ID (**not** SIM ID). For example: **T01F_JohnTan_9876543**.
- For part time student, the zip file name must be in the form of: TXXP_NAME_UOWID.zip where XX is your tutorial group, NAME is your full name (without space and underscore) and UOWID is your 7-digit UOW ID (**not** SIM ID). **T01P_JohnTan_9876543**.
- Wrong file naming (even an extra space, 5 marks will be deducted)
- Only zip file is accepted. Please do not use other compression format such as 7x, grip, rar and etc.
- Please ensure the UOW ID is correct as it will be used by the marking scripts to update your assignment mark in Moodle.
- Automated scripts will be used to extract your files for grading. Failure to follow the above instructions will result in failure to extract and grade your submission.
- No appeal will be granted for incorrect submission such as empty zip file, wrong files, etc.
- Late submission will be penalized 5% per day late.
- Late submission after 4 days will not be graded.

Auto Grading:

Another auto grading script (similar to A1_Assertion.java) will be used to grade your submission. Please ensure that you adhere to the given format strictly.

Do not change the function name and input and return parameters and data type.

Failure to execute your submission will results zero marks awarded.